Technology Specification Document

Khushboo Scholar Learning & Engagement Dashboard

Organization: Cybage – Khushboo Scholarship Program

Project Type: Hackathon Prototype / Academic Innovation Project

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Version: 1.0

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1. Purpose of This Document

This document provides a **detailed breakdown of system technology stack, and deployment architecture** for the Khushboo Scholar Learning & Engagement Dashboard. It complements the Product Requirement Document (PRD) by explaining **what the system will contain and how it will be built** at a technical and functional level.

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2. Technology Stack

Frontend

- **React.js**
- Responsive UI design
- Component-based architecture

Backend

- **Node.js**
- **Express.js** REST APIs
- Role-based access validation

Database

- **PostgreSQL**
- Structured relational schema

Database Platform

- **Supabase** (PostgreSQL hosting, authentication support)

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3. Deployment Architecture

Application Hosting

- **Render**
- Cloud-based deployment for frontend and backend

Database Hosting

- **Supabase**
- Secure PostgreSQL instance

Version Control

- **GitHub**
- Branch-based development workflow

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4. Security & Compliance

- Dummy data usage only
- Role-based access enforcement
- Secure API endpoints
- No real student or personal data stored

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5. Limitations

- Prototype-level scalability
- No third-party integrations
- No production deployment
- Limited analytics (hackathon scope)

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6. Conclusion

This Feature & Technology Specification document outlines a clear, professional, and hackathon-ready implementation approach for the Khushboo Scholar Learning & Engagement Dashboard. The selected technology stack and modular design align with modern web development practices while respecting Cybage CSR compliance requirements.